Problem 16.1

An 8-year annuity due has a present value of \$1,000. If the effective annual interest rate is 5%, then what is the value of the periodic payment?

Solution

Let X be the periodic payment. Since this is an annuity due, payments are made at the beginning of each year. Therefore, the present value is

$$1000 = X\ddot{a}_{\overline{8}|}.$$

For an annuity due,

$$\ddot{a}_{\overline{n}|} = (1+i)a_{\overline{n}|}.$$

Using $i = 0.05$,

$$\ddot{a}_{\overline{8}|} = (1.05)\left(\frac{1-(1.05)^{-8}}{0.05}\right).$$

Thus,

$$1000 = X(1.05)\left(\frac{1-(1.05)^{-8}}{0.05}\right).$$

Solving for X,

$$\begin{aligned} X &= \frac{1000}{(1.05)\left(\frac{1-(1.05)^{-8}}{0.05}\right)} \\ &\approx 147.3541. \end{aligned}$$

Hence, the periodic payment is

$$\boxed{\$147.35}.$$